

VR Project Design Document

1 App Info

Tentative Title:	VR Hockey Mayhem	
☒	Education & Training	☐ Mental Health & Fitness
☐	Travel & Discovery	☐ Media & Entertainment
☐	Productivity & Collaboration	☒ Gaming
☐	Art & Creativity	☐ Other: _____

2 Pitch

To goal is for users to [learn | experience | practice | review | design | play | other]:

To play a competitive, physically engaging air hockey game

This will be especially [impactful | educational | memorable | effective | fun | other] in VR b/c:

VR is impactful because it provides depth perception for tracking the puck in 3D space and allows the player to use physical arm movements to hit the puck

At a high level, during the app, users will:

Physically swing the mallet to strike the puck and try and score a goal

This experience will be targeted at devices with:

[3 6] 6	degrees of freedom, giving users control over the	[rotation movement & rotation] movement & rotation	of their head & controllers.
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3 Basics

The app will take place in:

Room with an air hockey table

and the user will get around the scene with:

[teleport stationary continuous other N/A - user will be stationary]	movement.
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The user will be able to grab:

-
- The air hockey mallet
-

There [will | will not] be sockets:

-
- Yes to ensure the mallet snaps to the
players hand position
-

4

Events & Interactions

By default, the left hand will have a:

Both [Direct Ray]	interactor.
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and the right hand will have a:

Both [Direct Ray]	interactor.
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And you [will | will not] be able to toggle on a [Direct | Ray] interactor using the joystick.
will ray

There will be haptic / audio feedback when:

- Vibration when the mallet hits the puck, and when scoring

There will also be 3D (diegetic) sound from:

- The crowd cheering behind the player, and the goal siren.

If the user is holding:

Either controller	and presses the trigger,	on the ui buttons it will click the button
	and presses the trigger,	
	and presses the trigger,	
		Suggestions: a UI change, a sound/video plays, a particle plays, an object is spawned or destroyed.

The main menu will be located:

Floating Panel

and from the main menu, the user will be able to:

- Start game, restart

[Optional] There will be additional UI elements for:

- -
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5

Optimization & Publishing

To make the user experience more accessible / comfortable:

- Possible air hockey table height adjustment -

Given that this app is targeting the [headset model], target metrics are:

Frames per second:	>= _____	FPS
Milliseconds per frame:	< _____	ms (= 1,000 / FPS)
Triangles per frame:	_____ - _____	tris
Draw calls per frame:	_____ - _____	batches

Lighting strategy:

☐ All baked	☒ Mostly baked with some mixed	☐ All real-time
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Light probes [will | will not] also be used for more realistic mixed lighting.

6
Other
features

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-

-

-

-

-

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7
Sketch
(Optional)

8
Timeline
(Optional)

	Milestone	Date
1	- Core XR Setup & Player Presence	6/2
2	- Physics-Based Interaction	8/2
3	- VR User Interface & Game Loop	10/2
4	- Audio & Environment Polish	12/2
5	- Accessibility for table height	14/2